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$$\text{At Q, } N + mg = \frac{mv^2}{r}$$

As the bead is just in contact with the wire, $N = 0$. Thus, $v^2 = rg = 0.20g$

Using conservation of energy,

$$mgh = \frac{1}{2}mv^2 + mg(0.40)$$

$$mgh = \frac{1}{2}m(0.20g) + mg(0.40)$$

$$h = 0.50 \text{ m}$$

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